

Effectiveness of a Generate–Verify–Repair Pipeline with Batfish for LLM-Generated Vendor CLI Configurations

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Abstract—We report a preregistered paired experiment (cnsms2026-001-gvr-batfish-prereg-v1) that evaluated whether a generate–verify–repair (GVR) pipeline—shared initial LLM candidate (gpt-5-mini) + a deterministic Batfish-style validator + at most one automated LLM repair—reduces deterministic-verifier detected semantic/safety failures on intent→vendor-CLI tasks. Planned N=300 pairs; 266 complete pairs were analysed after preregistered exclusions. Baseline pass-rate = 260/266 (97.744%); guarded (final GVR) pass-rate = 265/266 (99.624%). Paired difference = 0.01879699248; exact McNemar two-sided p = 0.0625. Paired bootstrap (B=10,000, seed=1729) 95% percentile CI = [0.0037593984962406013, 0.03759398496240601]. Repair was invoked for 6 tasks; median repair iterations per task = 0. Per-episode latency was not recorded in per-task artifacts and thus could not be evaluated. Under shared_initial_candidate semantics the observed absolute improvement is small and did not meet the preregistered $\alpha=0.05$ decision rule. We provide preregistered design details, deterministic validator semantics, exact paired counts, representative artifact-grounded repair examples, contamination findings (no exact public duplicates flagged), and a focused discussion of limitations and implications.

I. INTRODUCTION

Generative large language models (LLMs) are actively explored for NetOps tasks such as intent→CLI translation, zero-touch configuration synthesis, and automated maintenance workflows [1], and surveys emphasize both opportunity and risk when LLMs are applied to operational network configuration [2]. Stochastic LLM outputs can produce syntactic or semantic violations—missing required commands, unsafe command orderings, or constraint breaches—that create operational risk if applied directly. To mitigate that risk, pipelines that interpose deterministic verifiers (e.g., Batfish or header-space analyses) and bounded repair loops—collectively generate–verify–repair (GVR)—have been proposed to provide deterministic acceptance criteria and automated correction [3,4]. Prior work presents components and prototypes, but preregistered, paired evaluations that archive verifier traces and quantify verifier-triggered repair benefit on reproducible NetOps task banks remain limited.

This study tested the preregistered question (cnsms2026-001-gvr-batfish-prereg-v1): Does a GVR pipeline that uses a single sampled initial LLM candidate per task (shared_initial_candidate semantics), a deterministic Batfish-style validator, and at most one automated LLM repair call reduce deterministic-verifier detected semantic/safety failures relative to the same initial candidate without repair?

Our principal contribution is a preregistered, artifact-archived paired evaluation executed under the CNSM Agentic AI Researcher constraints that reports exact paired counts, inferential outcomes, execution accounting, representative artifact examples, and an evidence-grounded interpretation.

II. RELATED WORK

Prior work relevant to this study spans (a) applied LLM prototypes and benchmarks for network configuration, (b) surveys documenting LLM failure modes and recommendations to ground generation, (c) proposals for generate–verify–repair and related architectures for deterministic validation, and (d) established deterministic network verification tools that serve as practical oracles in automated pipelines. We synthesize those threads and emphasize aspects most relevant to a preregistered paired GVR evaluation.

LLM-driven NetOps prototypes and benchmarks: Recent system-level and benchmark efforts investigate whether LLMs can aid network configuration generation and automation. NetConfEval and similar task collections examine intent→CLI translation and measure correctness under vendor-dialect constraints; these works provide public task exemplars and motivate reproducible task generators for controlled evaluation [1]. Such benchmark tasks typically encode required command sequences and vendor syntax, which directly map to the validator check types we archive in this study (required-sequence, allowed-field checks, topology reachability). Importantly, prior benchmarks often evaluate single-pass generation accuracy rather than integrated verifier-triggered repair, which motivates the paired GVR design adopted here.

Failure modes, grounding, and mitigation recommendations: Survey and synthesis papers summarize that LLMs commonly produce both syntactic and semantic errors when applied to domain-constrained tasks: missing commands, incorrect ordering, and hallucinated or unsafe modifications [2]. Those surveys recommend grounding strategies (retrieval, tool use), constrained generation, and verifier insertion to reduce the operational risk of accepting model outputs without deterministic checks. Our work adopts those prescriptions by interposing a deterministic validator and bounding the repair loop to one automated correction attempt, aligning experimentally with the practical mitigations advocated in the literature.

Generate–verify–repair and bounded repair proposals: The idea of interposing deterministic oracles and using closed-loop

repair is well established in proposals for regulated or safety-sensitive domains. Convergent GVR architectures aim to combine stochastic generation with deterministic acceptance criteria and limited repair to achieve practical correctness guarantees while preserving the generative model’s productivity [3]. These proposals emphasize publishing verifier traces and repair prompts so that corrected outputs are auditable—an archival design we implemented in the execution artifacts (validator traces, `repair_prompt_sha256`, repair provider traces).

Deterministic network verifiers as oracles and their coverage limits: Tools like Batfish and header-space analyses provide deterministic, rule-based checks for reachability, ordering, and configuration invariants and have been incorporated into research prototypes that integrate model outputs with formal validation [4]. Batfish-style validators excel at expressing and mechanically checking reachability and required-sequence constraints on modeled topologies; however, their guarantees are strictly limited to the rule battery they encode. This distinction matters experimentally: a verifier that checks an explicit finite set of syntactic/semantic invariants yields a binary pass/fail endpoint that is reproducible and auditable, but it cannot detect semantic failures outside its coverage. Our deterministic `netops_validator_v5` follows this same engineering trade-off (syntactic parsing, required-sequence checks, allowed-fields enforcement, and topology reachability), and we archive per-episode validation `_before/after` traces so readers can inspect exactly which rules were exercised on each episode.

What LLMs need to synthesize correct router configurations and constrained decoding: Empirical studies analyzing LLM difficulty with router configuration point to sequencing, vendor-specific syntax, and dependency constraints as recurring challenges [5]. These analyses motivate two complementary mitigations: (1) stronger first-pass constraints or constrained decoding to reduce syntactic/format errors, and (2) verifier-triggered repair to correct semantic violations that slip through first pass. Our study deliberately fixes the initial candidate (`shared_initial_candidate`) and thereby isolates the marginal effect of the verifier-triggered repair step rather than conflating improvements due to constrained first-pass prompting or multiple draws.

Gaps in prior evaluations that motivated this study: The supplied literature shows active proposals and component evaluations but comparatively fewer preregistered, paired experiments that (i) use `shared_initial_candidate` semantics to measure the isolated effect of bounded repair, (ii) archive raw model traces together with deterministic validator traces and repair provider traces for each episode, and (iii) apply explicit exclusion, retry, and contamination rules in a confirmatory design. Those gaps motivated our preregistered paired design and the artifact-heavy archival choices (manifested in the execution and analysis artifacts we publish alongside this manuscript).

Methodologically relevant syntheses for experimental design: From the surveyed and prototyped prior work we derived three concrete design choices: (1) use a deterministic validator

battery (Batfish-style) to obtain an auditable binary endpoint; (2) bound the repair loop (≤ 1 automated repair call per failed candidate) to maintain a clear paired estimand under `shared_initial_candidate` semantics; and (3) preregister explicit missingness, retry, and contamination treatments to avoid post hoc analytic choices. Each of these choices has precedent or justification in the cited literature: verifiers as deterministic oracles [4], GVR architectures for regulated settings [3], and benchmark/task construction recommendations that emphasize reproducibility and contamination controls [1,2].

In sum, our related-work synthesis locates this study at the intersection of applied intent→CLI benchmarks, survey-driven mitigation recommendations, GVR methodological proposals, and deterministic verification tooling. The specific empirical contribution is a preregistered, archived paired GVR evaluation under `shared_initial_candidate` semantics that fills a documented gap in artifact-grounded confirmatory assessments of verifier-triggered repair effectiveness.

III. METHODOLOGY

Preregistration and estimand: The experiment followed the preregistered plan `cnsms2026-001-gvr-batfish-prereg-v1` (manifest SHA-256: `46a223492f760950b3d06f475129ce21e52e296c4c2a1e09df3ccd50bec9f891`). The primary estimand is the `paired_success_rate_difference_guarded_minus_baseline` under `shared_initial_candidate` semantics. Under these semantics the identical single sampled initial model candidate per task is used to produce both outcomes: (1) baseline — score the single initial candidate with the deterministic validator; (2) guarded — if that same candidate fails, the pipeline may invoke at most one automated LLM repair call and then rescore with the same deterministic validator. This design isolates the verifier-triggered repair effect for a fixed initial draw and matches the preregistered `execution_contract` (`generation_semantics = "shared_initial_candidate"`).

Task bank, sampling, and inclusion rules: The task bank was deterministically generated to produce $N=300$ intent→CLI pairs, stratified into two vendor-dialect clusters (150 Cisco-like, 150 Juniper-like) via a seeded synthetic generator supplemented with public NetConfEval-style items as specified in the preregistration. Generator ids and deterministic seeds are archived in the task manifest. Inclusion and exclusion rules followed the preregistered contract: public exact-duplicate tasks detected by the contamination check were to be excluded from the primary confirmatory analysis (none were excluded for exact duplication in this run); provider/API transient failures triggered up to two automatic retries, and pairs where both arms failed due to infrastructure/model API errors were excluded from the primary paired analysis per the `'complete_pair_primary'` missingness rule. The execution manifest records `planned_episode_count = 600` ($300 \text{ pairs} \times 2 \text{ arms}$) and `completed_episode_count` and exclusions are reconciled in the deterministic reconciliation artifacts.

Model orchestration and runtime configuration: The declared model used for generation and any repair calls was recorded as gpt-5-mini (execution/model_configuration.json declared_model_version = "gpt-5-mini"). Archived configuration fields include max_output_tokens = 1024, maximum_attempts_per_call = 1, provider = openai_responses, and temperature = null (unset). Provider accounting recorded hosted_model_call_event_count = 306, completed_hosted_model_call_event_count = 272, and failed_hosted_model_call_event_count = 34; stage accounting recorded shared_initial_generation = 300 and repair = 6. Because temperature was unset and no centralized deterministic sampling seed for provider calls is archived, nondeterministic sampling of provider responses cannot be excluded for the recorded calls; this runtime nondeterminism is declared and discussed in Limitations and the Disclosure Statement.

Deterministic validator design and scoring rules: The binary oracle was deterministic_netops_validator_v5. For each candidate it executed a fixed rule battery that is archived per episode: (1) CLI syntactic parsing and normalization; (2) required-sequence checks comparing the task's required_sequence to the parsed commands; (3) constrained-field touch checks against allowed_touched_fields and allowed_fields in the task payload; and (4) reachability/constraint validation over the deterministic synthetic topology model (a lightweight Batfish-style evaluation). Each validator trace archives validation_before and validation_after objects containing parsed_command_count, required_command_count, observed_touched_field_count, violation codes, normalized_configuration, and a boolean valid flag. The confirmatory scoring rule for the primary binary endpoint was 'full_intent_constraint_validation' as recorded in the scoring artifacts; these artifacts drive the paired contingency counts used in the primary test.

Repair loop mechanics and bounding: Per the preregistration the automated repair stage was strictly bounded to at most one LLM repair call per episode when the initial candidate failed the validator. The repair prompt construction is archived: it includes the task constraints, the validator failure codes and normalized validator diagnostics, and an explicit instruction to produce a corrected configuration satisfying the same task constraints. Repair provider traces and repair_prompt identifiers are archived for each repair invocation. Stage accounting indicates 6 repair calls executed in the run; all repair artifacts (repair prompts, repaired outputs, and subsequent validator traces) are included in the evidence bundle and illustrated in representative scoring JSONs.

Analysis plan and inferential procedures: The prespecified primary hypothesis test was the exact McNemar two-sided test on paired binary outcomes at $\alpha=0.05$ (estimator id paired_success_rate_difference_guarded_minus_baseline). Interval estimation used a paired bootstrap percentile CI (B=10,000, bootstrap_seed = 1729) for the paired difference to quantify effect magnitude and uncertainty; these bootstrap parameters and resample counts are archived in the

analysis log. Missingness handling for the primary analysis followed the preregistered 'complete_pair_primary' rule: pairs with both-arm provider/infrastructure failures were excluded from the confirmatory McNemar test; a preregistered sensitivity analysis treats excluded episodes conservatively as failures. Secondary analyses (preregistered) included median_repair_iterations_per_task and median end-to-end latency; the latter could not be computed because per-episode timing was not archived. Multiplicity control for formal secondary tests was specified with a Bonferroni adjustment; exploratory analyses were explicitly labeled and not used to support the primary claim.

Execution logging, archival, and auditability: Every episode archived raw model outputs, normalized configurations, validator_before/after traces, scorer decision codes, and provider call accounting. Deterministic reconciliation artifacts verify episode accounting (completed_episode_count, failed_episode_count, complete_pair_count) and provider call tallies. Key execution statistics recorded in the reconciliation artifacts are: planned_episode_count = 600, completed_episode_count = 532, failed_episode_count = 68, complete_pair_count = 266, model_calls_used = 306, repair_calls = 6, cache_hit_count = 0, cache_miss_count = 272. These archived artifacts permit independent verification of the paired counts (n11, n10, n01, n00) and replay of repaired episodes for failure-mode inspection.

Design constraints and estimand implications: We emphasize that shared_initial_candidate semantics (one sampled initial candidate per task used to evaluate both baseline and guarded outcomes) was selected and executed exactly as preregistered. This choice isolates the incremental effect of a validator-triggered, bounded repair for the observed initial sample; it does not measure the effect of constrained prompting, ensemble first-pass generation, or multiple independent initial draws. Those alternative interventions would require independent-generation arms or additional preregistered conditions that were not executed under this run and therefore are outside the estimand and scope of the reported confirmatory analysis.

IV. RESULTS

Execution accounting and missingness: Planned episodes = 600; completed episodes = 532; failed episodes = 68 (execution_manifest.json). After applying preregistered exclusion rules (both-arm provider/infrastructure failures), the complete paired units analysed = 266 (analysis/missingness_summary.csv). Provider and model call accounting (deterministic_reconciliation.json) reports hosted_model_call_event_count = 306, completed_hosted_model_call_event_count = 272, failed_hosted_model_call_event_count = 34, cache_hit_count = 0, cache_miss_count = 272, and stage_counts showing shared_initial_generation = 300 and repair = 6.

Primary confirmatory outcomes (complete_pairs = 266): baseline successes = 260; guarded final successes = 265. Paired contingency counts

(analysis/paired_contingency_table.csv) are $n_{11} = 260$ (both pass), $n_{10} = 5$ (guarded pass only), $n_{01} = 0$ (baseline pass only), $n_{00} = 1$ (both fail). Observed proportions: baseline = $260/266 = 0.9774436090$; guarded = $265/266 = 0.9962406015$. Paired difference = 0.01879699248 . Exact McNemar two-sided $p = 0.0625$ (preregistered $\alpha=0.05$ decision rule $\rightarrow$ not significant). Paired bootstrap percentile CI ($B=10,000$, seed=1729) = $[0.0037593984962406013, 0.03759398496240601]$ (analysis_log.jsonl archives bootstrap parameters and seed). The differing conclusions between exact McNemar ($p>0.05$) and the bootstrap CI (excludes zero) reflect small discordant counts and discrete p-value behavior: with few discordants (5 vs 0) exact McNemar’s two-sided discrete p can be conservative while bootstrap CI summarizes resampled paired-difference variability.

Repair behavior and representative artifact examples: Repair stage calls recorded = 6; median repair iterations per task = 0. Representative episode: task-000008 shared initial candidate (execution/responses/task-000008-shared-initial.txt, sha256 = 89306095...) contained a truncated final token ("interface uplink2\n") and failed validation_before (parsed_command_count mismatch). The guarded final artifact (execution/responses/task-000008-guarded.txt, sha256 = d52b7616f...) contained the corrected sequence including the previously truncated line and the required edge1 commands; its validator trace (execution/scoring/task-000008-guarded.json) shows validation_after with valid = true and no violation codes. All repair invocations record repair_prompt_sha256 and repair_provider_trace_path in the archived scoring artifacts; these traces include the validator failure codes used by the repair prompt and the repaired configuration accepted by the validator.

Contamination and sensitivity analyses: The contamination detector (analysis/contamination_summary.csv) produced no exact public-duplicate flags; therefore no exact-duplicate tasks were excluded for contamination in the primary analysis. A preregistered sensitivity analysis that treats the 34 excluded both-arm episodes as failures (conservative complete $N=300$) yields baseline = $260/300 = 0.866667$ and guarded = $265/300 = 0.883333$ but retains discordant counts $n_{10}=5$, $n_{01}=0$ and exact McNemar $p = 0.0625$, indicating the primary inferential conclusion is robust to the preregistered conservative missingness handling.

Supplementary quantitative artifacts: analysis/condition_summary.csv and analysis/paired_contingency_table.csv are archived and hashed. deterministic_reconciliation.json documents that all deterministic consistency checks passed and provides provider call and stage counts used above. Per-episode validator traces (execution/scoring/*.json) provide normalized_configuration and explicit violation codes for every episode and permit replay or reanalysis of individual failure modes.

V. DISCUSSION

Statistical interpretation and small-discordant behavior: The guarded GVR pipeline produced a small absolute improve-

ment (~ 1.88 percentage points). Under the preregistered exact McNemar decision rule the improvement did not reach $\alpha=0.05$ ($p=0.0625$). The paired bootstrap CI that excludes zero highlights finite-sample discreteness: with few discordants ($n_{10}=5$, $n_{01}=0$) McNemar’s discrete two-sided p-value can be conservative for small effects while bootstrap percentile CIs reflect resampled paired-difference variability. We report both inferential perspectives so readers may weigh formal hypothesis rejection against effect magnitude and CI width.

Operational implications and boundary conditions grounded in artifacts: Baseline first-pass accuracy was high ($\approx 97.7\%$), leaving few failures for the repair stage to correct—hence rare repair invocations (6/300). Stage accounting and provider traces show repair-stage overhead was small in the observed run, but we did not archive per-episode latency and therefore cannot quantify repair latency or end-to-end timing against the preregistered operational bound (median ≤ 60 s). The low repair rate suggests bounded automated repair adds limited marginal benefit on high-first-pass corpora; however, GVR may yield larger absolute improvements when first-pass accuracy is lower, verifier coverage expands, or when the verifier detects failure modes common in production workloads.

Threats to validity (artifact-grounded): Internal validity: the shared_initial_candidate estimand measures the effect of validator-triggered repair for a fixed initial sample and does not assess independent multiple draws or constrained first-pass prompting because independent first-pass arms were not executed under the preregistration. Construct validity: deterministic_netops_validator_v5 enforces a finite, archived rule battery (syntactic parsing, required_sequence, constrained_field checks, reachability) and therefore semantic violations outside that battery are undetected and unmeasured. External validity: tasks derive from public NetConfEval-style examples and a seeded synthetic generator limited to two vendor dialects; results may not generalize to other vendors, larger topologies, or production heterogeneous workloads. Conclusion validity: the loss of 34 both-arm provider failures reduced effective N and the observed few discordant pairs limit power for small effect detection.

Design lessons and recommendations grounded in execution artifacts: (1) When baseline pass rates are high, design experiments targeting lower-baseline corpora or include arms that modify first-pass sampling (e.g., multiple draws or constrained prompts) to increase discordant counts and power. (2) Archive per-episode pipeline timing (request $\rightarrow$ generation $\rightarrow$ validation $\rightarrow$ repair $\rightarrow$ final validation) to assess operational latency bounds; our run lacked per-episode latency and thus the preregistered latency estimand could not be evaluated. (3) Expand and publish validator rule coverage and include tests that exercise verifier false negatives; per-episode validator traces permit such audits and should be part of future preregistrations. (4) Consider paired designs that explicitly compare shared_initial_candidate semantics to independent generation semantics in separate preregistered arms; such contrasts require independent initial draws and were outside this run by design.

Reproducibility and archival resources: All raw responses, execution/provider call traces, validator traces, scoring JSONs, analysis artifacts (paired contingency table, condition summary, missingness summary, contamination summary), `deterministic_reconciliation.json`, and `model_configuration.json` are archived and hashed in the evidence bundle. The archived `analysis_log.jsonl` records bootstrap settings ($B=10,000$, $\text{seed}=1729$). These artifacts permit exact reexecution of the confirmatory analysis and targeted replay of repaired episodes.

VI. LIMITATIONS

- `Shared_initial_candidate` semantics: both arms used the same single sampled initial candidate per preregistration; the estimand measures validator-triggered repair benefit for that fixed initial sample and does not evaluate independent first-pass generations or constrained first-pass prompting strategies.
- Missingness and sample reduction: planned $N=300$ pairs; after infrastructure/provider exclusions 266 pairs were complete and 34 both-arm episodes were excluded per preregistered rules, reducing effective sample size and precision.
- Small discordant counts: primary inference used exact McNemar with few discordants ($n_{10}=5$, $n_{01}=0$); conclusions are sensitive to inferential choice and limited power.
- Validator coverage: `deterministic_netops_validator_v5` enforces a finite set of syntactic/semantic/reachability rules; semantic violations outside its coverage are possible and unmeasured.
- Runtime nondeterminism and sampling: archived `execution/model_configuration.json` records `temperature = null` and no deterministic seed; nondeterministic sampling cannot be excluded for recorded model calls.
- H2 latency not evaluated: per-episode end-to-end latency was not present in archived per-task artifacts and therefore the preregistered latency bound (median ≤ 60 s) could not be assessed.
- External validity: task corpus derives from public NetConfEval-style examples and a seeded synthetic generator limited to two vendor-dialect clusters; results do not imply universal benefit across other vendors, larger topologies, or production deployments.

VII. CONCLUSION

We executed a preregistered paired experiment that evaluated a GVR pipeline (`gpt-5-mini` + `deterministic_netops_validator_v5` + ≤ 1 automated repair) on `intent`→`CLI` tasks under `shared_initial_candidate` semantics. Across 266 analysed pairs the guarded pipeline improved validator pass rate by 0.01879699248 but did not meet the preregistered exact-McNemar $\alpha=0.05$ decision rule ($p=0.0625$); a paired bootstrap CI ($B=10,000$, $\text{seed}=1729$) narrowly excludes zero. Repair calls were rare (6/300) and median repair iterations per task = 0. Per-episode latency was not archived and could not be assessed. All execution and analysis artifacts (raw responses, validator traces, provider

call accounting, and reconciliation manifests) are archived in the evidence bundle to support reanalysis and follow-on work.

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DISCLOSURE STATEMENT

Disclosure (concise): Declared model: `gpt-5-mini` (`execution/model_configuration.json` archived; `declared_model_version = "gpt-5-mini"`). Orchestration adapter: `hosted_netops_gvr_v1`. Deterministic validator: `deterministic_netops_validator_v5`. Preregistration: `cnsms2026-001-gvr-batfish-prereg-v1` (manifest SHA-256: 46a223492f760950b3d06f475129ce21e52e296c4c2a1e09df3ccd50bec9f891). Initial master prompt immutable reference: `provenance/master_prompt.txt` (SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b39b15ba). Human role and autonomy boundary: before the autonomy lock a human Research Director selected the topic family and approved pre-lock infrastructure development; after the autonomy lock the registered pipeline autonomously performed literature selection, preregistration, experimental execution, analysis, manuscript drafting, AI peer review simulation, and revision. Nondeterminism disclosure: archived `execution/model_configuration.json` records `temperature = null` (unset) and no deterministic sampling seed is archived; therefore nondeterministic sampling of model outputs cannot be excluded for the recorded provider calls. Execution and analysis artifacts, logs, and hash manifests are archived in the evidence bundle referenced in the preregistration.

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